



STABILIA 2K26

Where Balance Becomes Power

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STABILIA – BALANCING ROBOT CHALLENGE

1. Introduction

STABILIA – Balancing Robot Challenge is a technical robotics competition designed to test the **stability, control, and performance** of a two-wheel self-balancing robot. The competition consists of **three progressive rounds**, moving from pure balance verification to control-based racing and real-world challenges.

The event is open to all participants and encourages innovation, mechanical design skills, and control system knowledge.

2. Robot Specifications

2.1 Robot Type

- Two-wheel **self-balancing robot** only
- Balance must be achieved **using sensors and control algorithms**
- No additional support wheels, legs, or sliders are allowed

2.2 Dimensions (Maximum Allowed)

- Length : **20 cm -28 cm (Wheel to Wheel Distance)**
- Width: **10-12 cm**
- Height : **Below 20 cm (From ground)**

The robot must fit within the above dimensions at all times during the competition.

2.3 Weight

- Maximum weight: **2.5 kg** (including battery)

2.4 Wheels & Motors

- Exactly **2 wheels** are permitted

- Wheel diameter: **65–75 mm (recommended)**
 - Both wheels must be identical in size and type
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3. Power & Electronics Rules

3.1 Power Supply

- Battery type: Li-ion / Li-Po
- Maximum voltage: **6V to 9V (2S)**

3.2 Controller

- Any microcontroller is allowed (Arduino, ESP32, STM32 Blue Pill etc.)
- Sensor : MPU6050 (gyroscope + Accelerometer)

3.3 Safety

- All wiring must be properly insulated
 - Emergency stop (hardware or software) is **strongly recommended**
 - Judges may disqualify unsafe robots
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❖ Detailed Components List

1. Control & Processing Unit

Microcontroller Board (any one):

- ESP32 Dev Module (recommended – Wi-Fi + Bluetooth built-in)
 - Arduino Uno / Nano (with external Bluetooth module)
 - STM32 (Blue Pill)
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2. Sensors (any one):

- IMU Sensor
- MPU6050 / MPU9250

Used for tilt angle, acceleration, and balancing control

3. Motors & Motion

DC Gear Motors or BO Motors – 2 Nos

- RPM: 150–200 RPM
- Voltage: 6–9V

Motor Driver Module (any one):

- L298N / L293D / BTS7960

Wheels – 2 Nos

- Diameter: 65–75 mm
 - Both wheels must be identical
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4. Power Supply

- Battery : Li-ion / Li-Po Battery
- Maximum voltage: 9V (2S)

- Battery Holder / Pack
 - Power Switch
 - Buck Converter (Recommended)
 - To step down voltage to 5V / 3.3V for controller
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5. Mobile Communication (Semi-Automatic Rounds)

Option A – Bluetooth Control

- HC-05 / HC-06 Bluetooth Module
(if Arduino is used)

Option B – Wi-Fi Control

- ESP32 Built-in Wi-Fi
- Mobile phone connects directly via Wi-Fi
- No internet required

Mobile Control Applications (Examples):

- Bluetooth RC Controller
- ESP32 WiFi Robot App
- Custom Android App / Web Interface

Only Forward, Backward, Left, Right, Stop commands are allowed.

6. Mechanical & Structural Parts

- Chassis : 3D-printed / Aluminium / Acrylic
- Motor Mounts
- Battery Tray
- Spacers & Standoffs
- Couplers / Wheel Hubs

7. Wiring & Accessories

- Jumper Wires (Male–Male, Male–Female)
 - Screw terminals
 - Cable ties
 - Heat shrink tubes
 - PCB / Perf board (optional)
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4. Competition Format

The competition consists of **three rounds**. Each round must be cleared to advance to the next.

5. ROUND 1 – STABILITY ROUND (Fully Automatic)

5.1 Objective

To evaluate the **pure balancing capability** of the robot.

5.2 Control Type

- **Fully Automatic**
- No mobile control or human input after start

5.3 Task Description

- The robot must balance continuously for **30-40 seconds**
- The robot must remain within a **Given area**
- Forward or backward motion is not required

5.4 Attempts

- Maximum **3 attempts** allowed

5.5 Scoring (100 Marks)

Criteria	Marks
Continuous balance (30-40 sec)	60
Smoothness & stability	20
Design & safety	20

6. ROUND 2 – CONTROL & ENDURANCE ROUND (Semi-Automatic)

6.1 Objective

To test **automatic balancing** combined with **manual movement control**.

6.2 Control Type

- Balance: Automatic
- Movement: Manual (mobile control)

Allowed movements: - Forward - Backward - Left - Right - Stop

6.3 Track Details

- Track length: **2 meters (straight)**
- Track width: **35 cm**
- Surface: Flat

6.4 Task Description

1. Start from Point A
2. Reach Point B (2 meters)
3. Return to Point A
4. Robot must balance throughout the run

6.5 Scoring (100 Marks)

Criteria	Marks
Track completion	40
Time performance	40
Stability (no fall)	20

Fastest completion time scores highest.

7. ROUND 3 – CHALLENGE RACE ROUND (Semi-Automatic)

7.1 Objective

To evaluate robot performance under **real-world challenges** in a **head-to-head race**.

7.2 Control Type

- Semi-Automatic (same as Round 2)

7.3 Track Elements

- Small slope (5–8°)
- Gentle turns
- Speed zone
- Balance zone (robot must stop for 3 seconds)

No jumps or sharp turns will be included.

7.4 Race Format

- Two robots compete at a time
- Knockout elimination

Sr. No.	Evaluation Criteria	Description	Max Marks
1	Race Result	Winner of head-to-head race	50
		Runner-up of the race	30
2	Slope Completion	Successfully clears slope section	10
3	Turn / Curve Section	Successfully clears turn without fall	10
4	Balance / Stop Zone	Robot stops & balances for 5 seconds	10
5	Stability & Control	Smooth balancing, no jerks or oscillations	20
		TOTAL	100

8. General Rules & Penalties

- Intentional damage to track or opponent: **Immediate disqualification**
 - Judges' decision is final and binding
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9. Code & Fair Play Rules

- Teams must use **original code**
 - Wireless interference with other robots is prohibited
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10. Final Notes

- Organizers reserve the right to modify rules if required for safety or fairness
 - Any changes will be communicated before the competition
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STABILIA – Balancing Robot Challenge aims to promote hands-on learning, innovation, and engineering excellence.

We wish all participants the very best!